



TARGET-90

JKBOSE CLASS-11TH

Plant Kingdom

One Shot

BIOLOGY

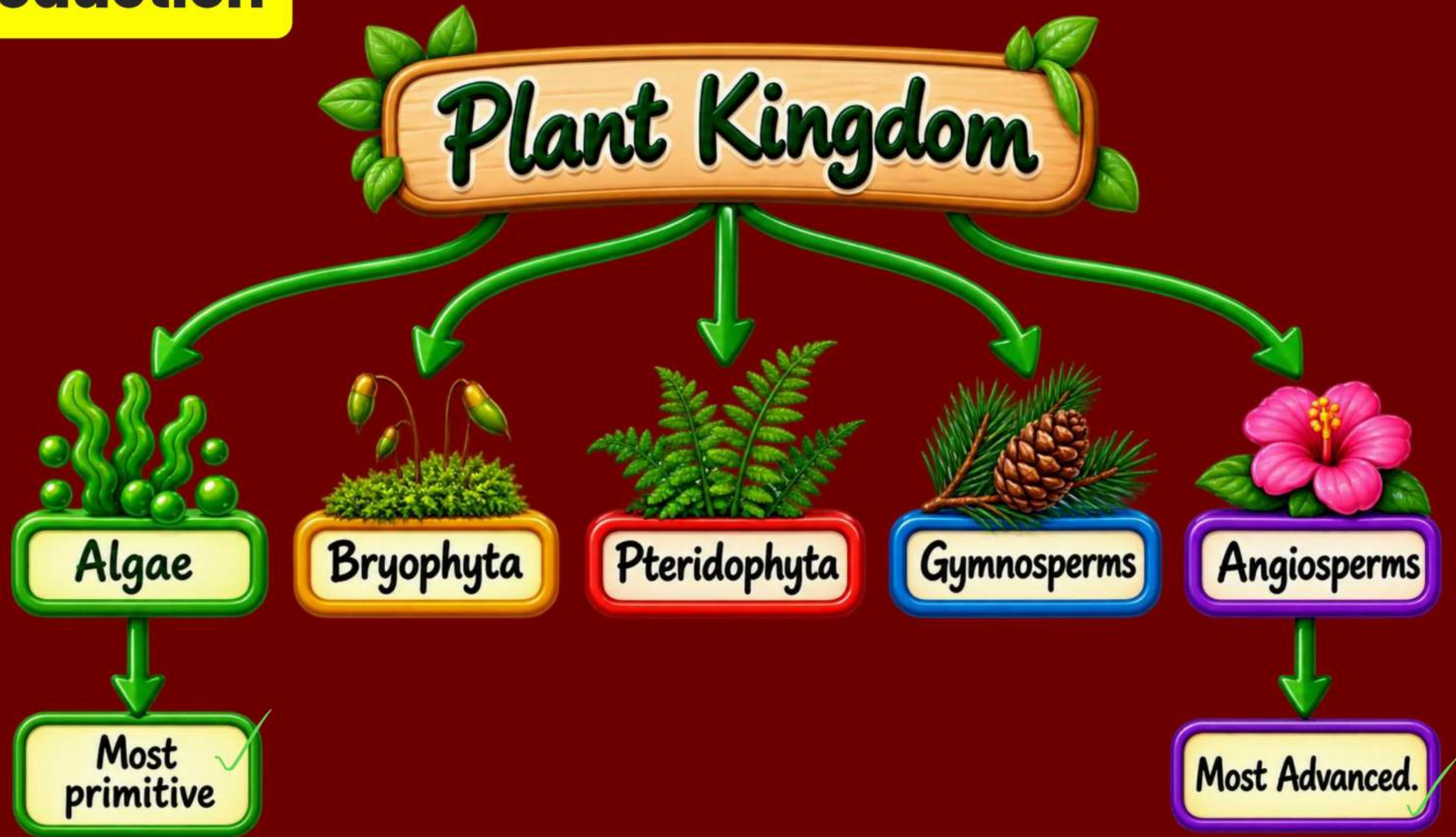
ARSHID Sir



Topics To Be Covered

- 1 Most Important Concepts (MIC)
- 2 ...

Introduction



Branches of Taxonomy

- 1) **Chemotaxonomy:** Plants are classified on the basis of biochemical characters like alkaloids, tannins, nucleic acids etc.
- 2) **Cytotaxonomy:** Organisms are classified on the basis of cytological characters like chromosome number, chromosome structure, etc.
- 3) **Numerical taxonomy:** Organisms are classified on the basis of all the observable characters.
 - Computers are used ✓
 - Number and codes are assigned to all the characters & the data are then processed.

Plant Classification — Artificial Classification

(1) Artificial Classification: Earliest system of classification.

Criteria → Gross morphological characters like habit, no. & shape of leaves etc.

Example: Theoprastus (Father of Botany)

Plants (on the basis of habit) — 04 groups

1. Herbs ✓
2. Under Shrubs ✓
3. Shrubs ✓
4. Trees ✓

Plant Classification — Artificial Classification

DRAWBACK:

1. Equal weightage given to vegetative & reproductive parts.
2. No use of phylogeny.
↓
(Evolutionary relationship among organisms.)

Natural Classification

(2) Natural Classification:

Criteria used: Morphological characters, Anatomical characters, Embryological, Phytochemistry, Cytological characters etc.

Example: G. Bentham & J.D Hooker.

↓
Classified **Spermatophytes** (seed bearing plants) into :

- Angiosperms
- 1. Dicots
 - 2. Gymnosperms
 - 3. Monocots

Drawbacks of Natural Classification

Drawbacks:

- a) No use of phylogeny. ✓
- b) Placement of Gymnosperms b/w Dicots & Monocots — Not acceptable. ✓

(3) Phylogenetic Classification:

AKA Cladistics. ✓

Criteria:

- Evolutionary relationship among organisms. ✓
- Morphological, Anatomical characters etc. ✓
- Most acceptable classification. ✓

General Characters of Algae — Habitat & Plant Body

Habitat: Aquatic

- Fresh H_2O (Salt Conc. ↓↓)
- Marine H_2O (Salt Conc. ↑↑)
- Brakish H_2O (Salt Conc. intermediate)

- Some algae lives in close association with fungi → 'LICHEN'.
- Some form close association with animals eg. on Sloth bear.

Plant Body: Thalloid (Not differentiated into Root, Stem, leaf).

Haploid (n) → Gametophyte → produces Gametes(n) by mitosis.

Vascular System: ABSENT

General Characters of Algae — Habitat & Plant Body

Size: Vary greatly. ✓

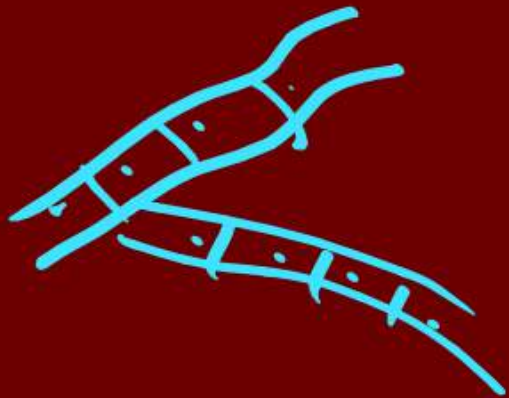
Body forms:

Unicellular

- Motile Ex. Chlymadomonos
- Non-motile Ex. Chlorella

Multicellular

- Colonial Ex. Volvox
- Filamentous Ex. Spirogyra, Ulothrix.



Reproduction in Algae

Reproduction: 03 Types

1. Vegetative rep.: By Fragmentation.

Ex. All filamentous algae.

2. Asexual rep.: By diff. types of Spores.

Ex. Zoospore — most common, Motile

3. Sexual rep.: → Fusion of gametes.

Male sex organ → Anthredium

Female sex organ → Oogonium

Sex organs — Unicellular & Jacketless

Jacket — Group of sterile cells covers sex organs.

3 Types of Sexual Reproduction

a) **Isogamy**: Fusing gametes → Morphologically similar

Ex. Ulothrix, Spirogyra.

b) **Anisogamy**: Fusing gametes → Morphologically dissimilar

Ex. Eudorina.

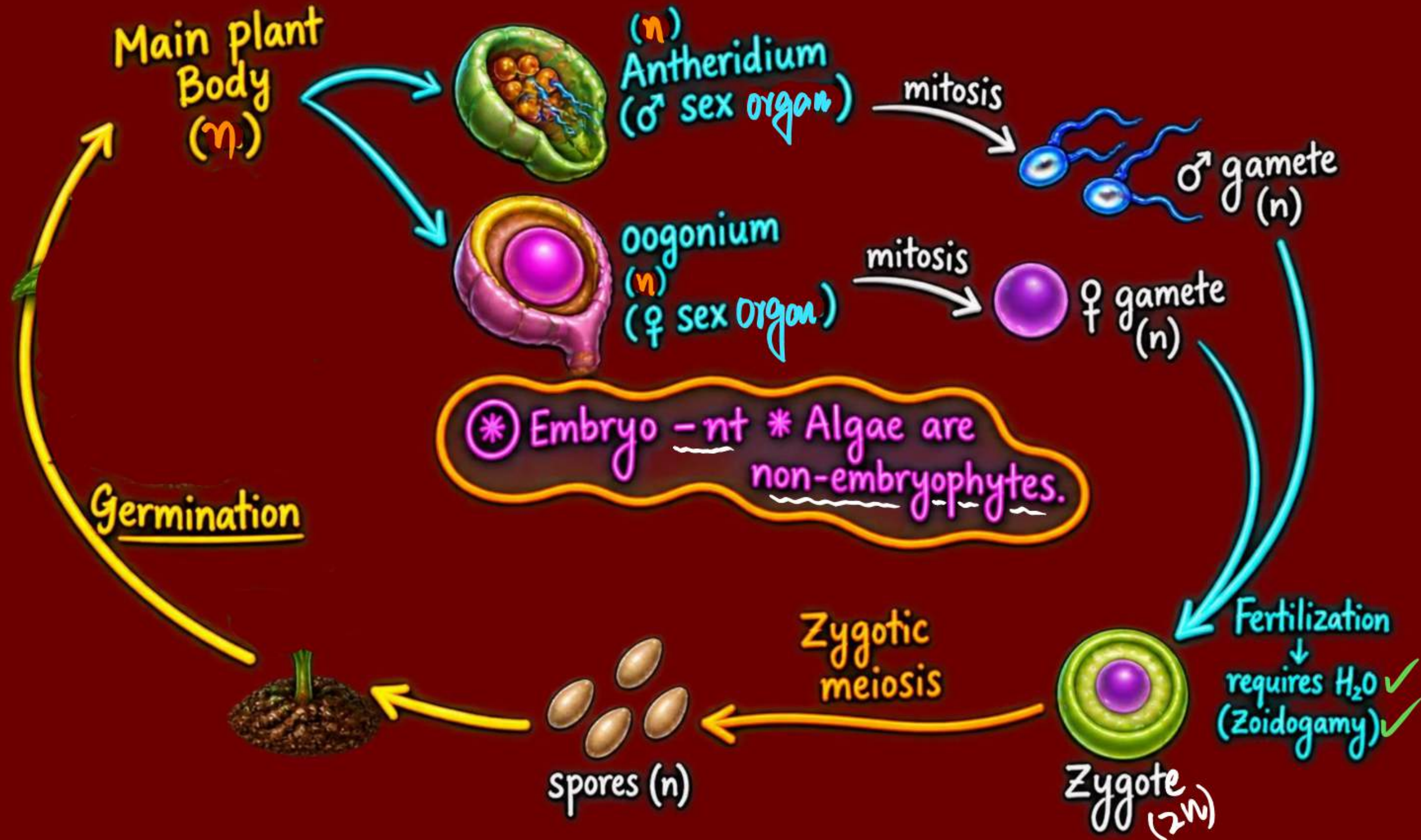
c) **Oogamy**: (Advanced)

Fusing gametes → dissimilar

- One small and flagellated → ♂ gamete 
- 2nd large & non-motile → ♀ gamete 

Ex. Volvox, Fucus.

Life Cycle Pattern in Algae



Classification of Algae

On the basis of:

- a) Photosynthetic pigment. ✓
- b) Cell wall composition. ✓
- c) Stored food material. ✓

Algae divided into 3 classes:

- (1) Chlorophyceae (Green algae)
- (2) Pheophyceae (Brown algae)
- (3) Rhodophyceae (Red algae).

Characters	Chlorophyceae (Green algae)	Pheophyceae (Brown algae)	Rhodophyceae (Red algae)
① Habitat	FRESH H ₂ O MARINE H ₂ O BRACKISH H ₂ O	MARINE H ₂ O (Maximum)	Marine H ₂ O (max.)
Flagella	2-8 in no. Apical in position Equal size 	2 in no. Laterol in position Unequal size 	ABSENT
Pigment	<u>Chl-a, Chl-b</u> (provides <u>green</u> <u>grass colour</u>) <u>Carotene, Xanthophyll</u>	<u>Chl-a, Chl-C, Carotene</u> <u>Fucoxanthin</u> (provides colour)	<u>Chl-a, Chl-d, Carotene</u> <u>γ-Phycoerythrin</u> (provides colour)

Imp

Imp

Characters	Chlorophyceae	Pheophyceae	Rhodophyceae
Cell wall 	Inner layer → Made of <u>Cellulose</u> Outer layer → Made of <u>Pectin</u>	Inner layer → Made of <u>Cellulose</u> Outer layer → Made of <u>Algin</u> (hydrocolloid i.e. H ₂ O holding substance)	Cellulose, <u>Pectin</u> , <u>Polysulphate</u> esters Outer covering made of <u>Carrageen</u> (hydrocolloid)

Characters	Chlorophyceae	Pheophyceae	Rhodophyceae
imp Stored food	<u>Starch</u> in <u>Pyrenoid Bodies</u> (Pyrenoid Body: Starch, Protein core inside chloroplast) 	<u>Laminarin &</u> <u>Mannitol</u> Complex Carbohydrates 	<u>Floridean Starch</u>
Examples	Chlamydomonas, Volvox, Spirogyra, Ulothrix, Chlorella	Laminaria, Dictyota, Ectocarpus, Kelp (Giant algae)	Gelidium, Gracillaria, Polysiphonia, Porphyra

Economic Importance of Algae

- ① More than 50% of CO_2 is fixed by Algae
Produces O_2 in their immediate environment

ecological importance.

②

Laminaria, Sargassum

- Brown algae

Porphyra

- Red algae

→ Edible

→ used as sea food.

- ③ Hydrocolloids — Algin & Carrageen } Commercial importance

- ④ Agar — obtained from Gelidium & Gracillaria } Red algae

→ Solidifying agent

→ Used in jellies & ice-creams

Economic Importance of Algae

⑤ Chlorella ↴

Unicellular green algae

Highly proteinitious

Used by space travellers as source of food & oxygen

General Characters of Bryophytes

Habitat: Commonly growing in moist shaded areas in the hills.

Main plant body: Haploid (n) → Gametophyte → produce gametes by mitosis.

Thalloid (more differentiated than algal thallus)

Vascular system — Absent

Possess:

- Root like structures (Rhizoids) — Unicellular/Multicellular ✓
- Stem like structure (Cauloid) — Prostrate/Erect  ✓
- Leaf like structures (Phylloid) 

 **Bryophytes** — lives on land but require H_2O for fertilization — Amphibians of
Plant Kingdom

General Characters of Bryophytes

- Sex organs — Multicellular and jacketed
- Male sex organ — Antheridium
- Female sex organ — Archegonium (Flask shaped) 
- Sexual reproduction — Oogamous
- Same kind of spores are formed (Homosporous)
- Embryo is present → Bryophytes are primitive embryophytes

Classification of Bryophytes

Bryophytes are classified into 03 classes:

- a) Liverworts (Hepaticopsida)
 - b) Hornworts (Anthocerotopsida)
 - c) Mosses (Bryopsida)
-] Not in Syllabus.

Liverworts



Ex. Marchantia

Rhizoids – Unicellular (present on 'Ventral Surface' of thallus).

Dorsal Surface – Possess Gemma Cups & bears Asexual Buds known as Gemma Buds.

Gemma Buds:

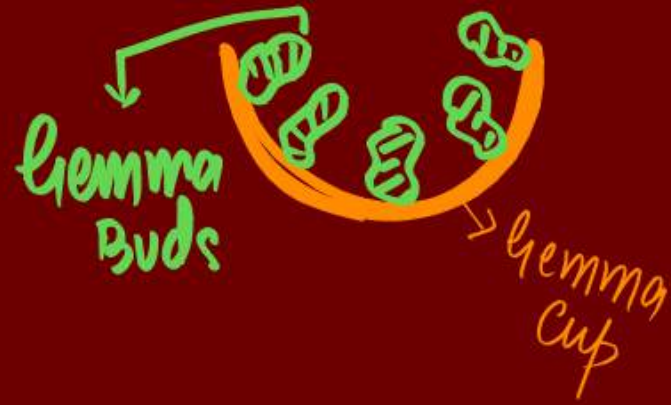
- ✓ Multicellular
- ✓ GREEN
- ✓ Helps in asexual reproduction.

Plant Body — Prostrate. 

Marchantia is Dioecious i.e., male & female sex organs are present on two different thallus.

lower surface of thallus.

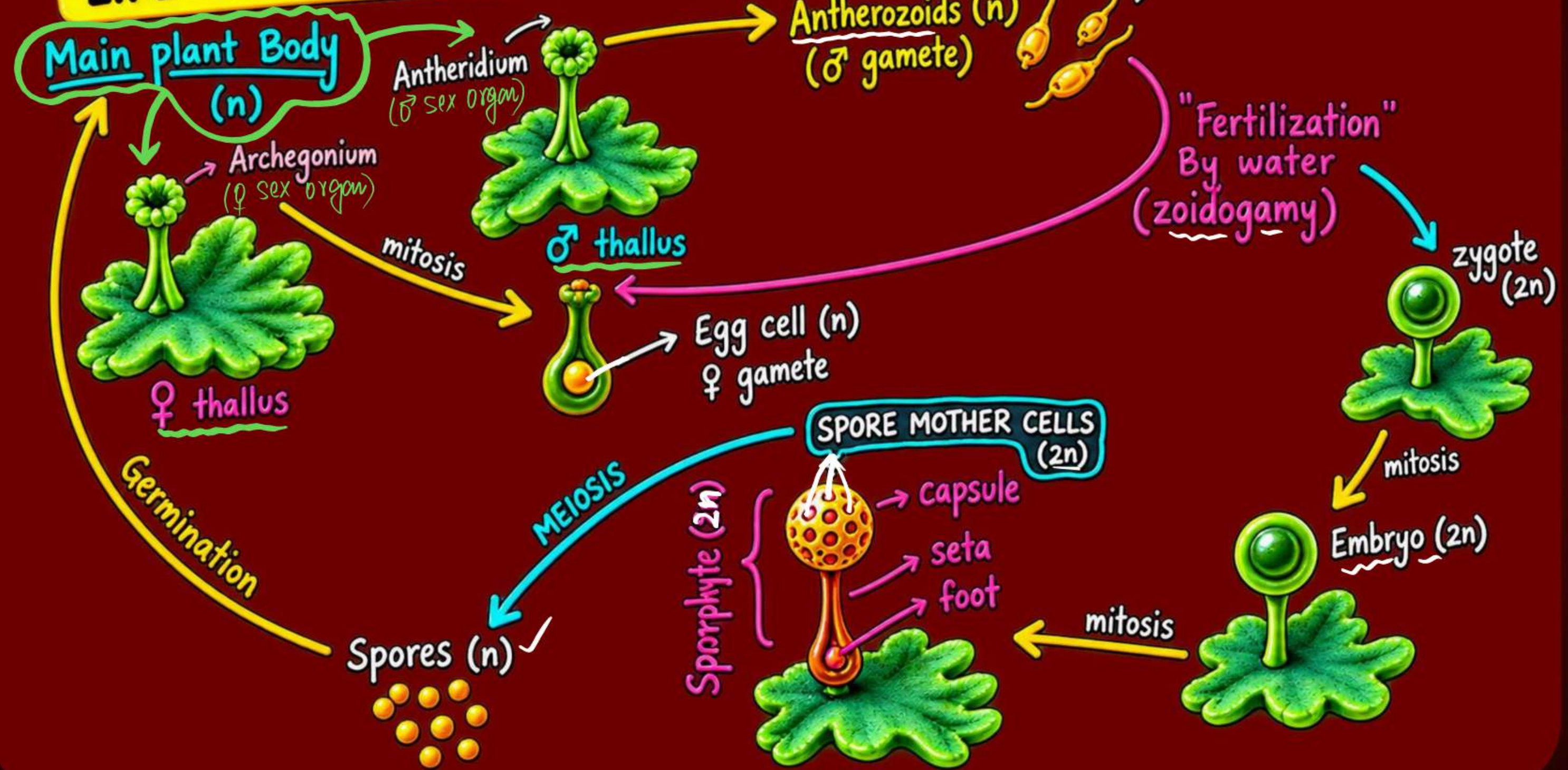
upper surface



Gemma Cups

LIFE CYCLE OF LIVERWORTS

Sporophyte - $2n$ str. which produces spores(n) by meiosis



MOSSES

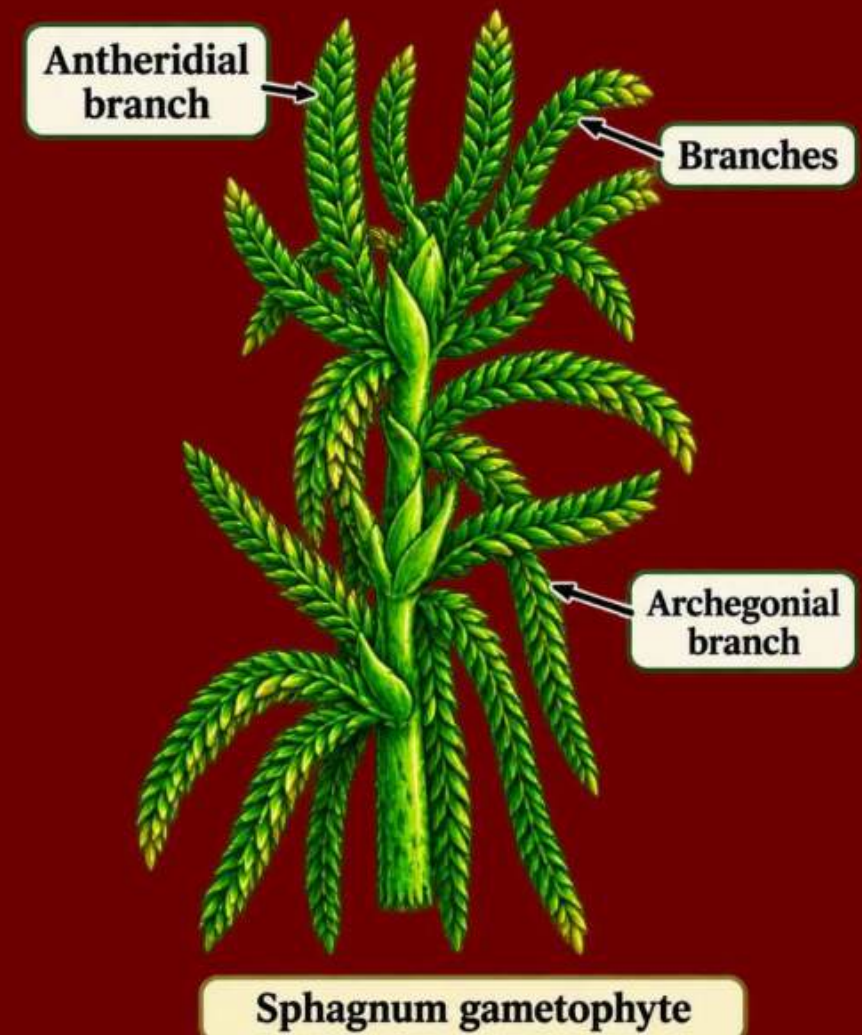
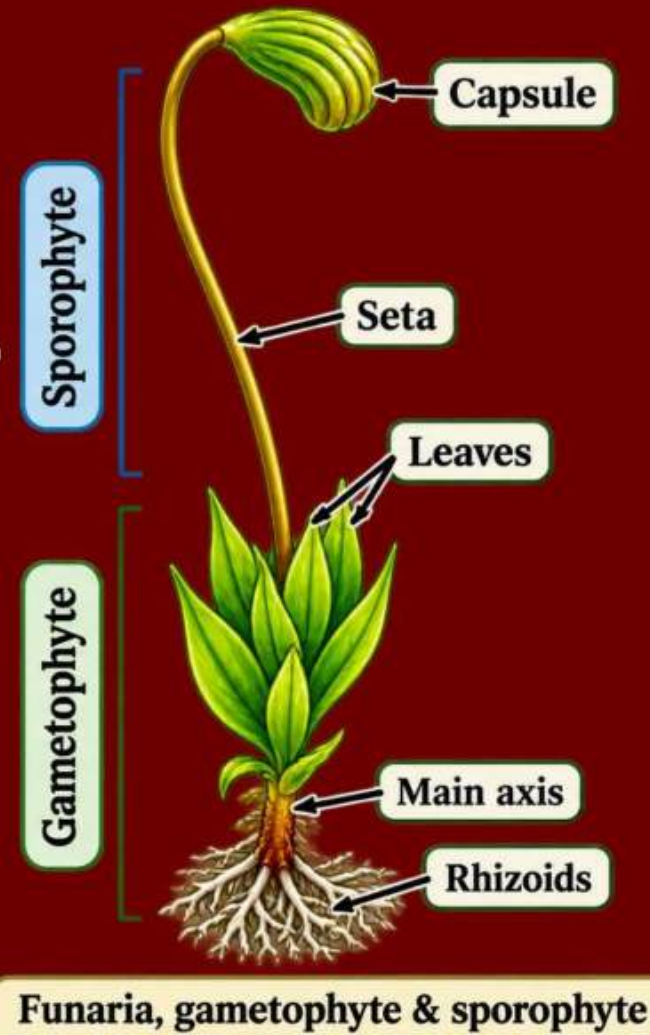
Rhizoids — multicellular

Plant Body — Erect.

Ex. Funaria, Sphagnum, Polytrichum.

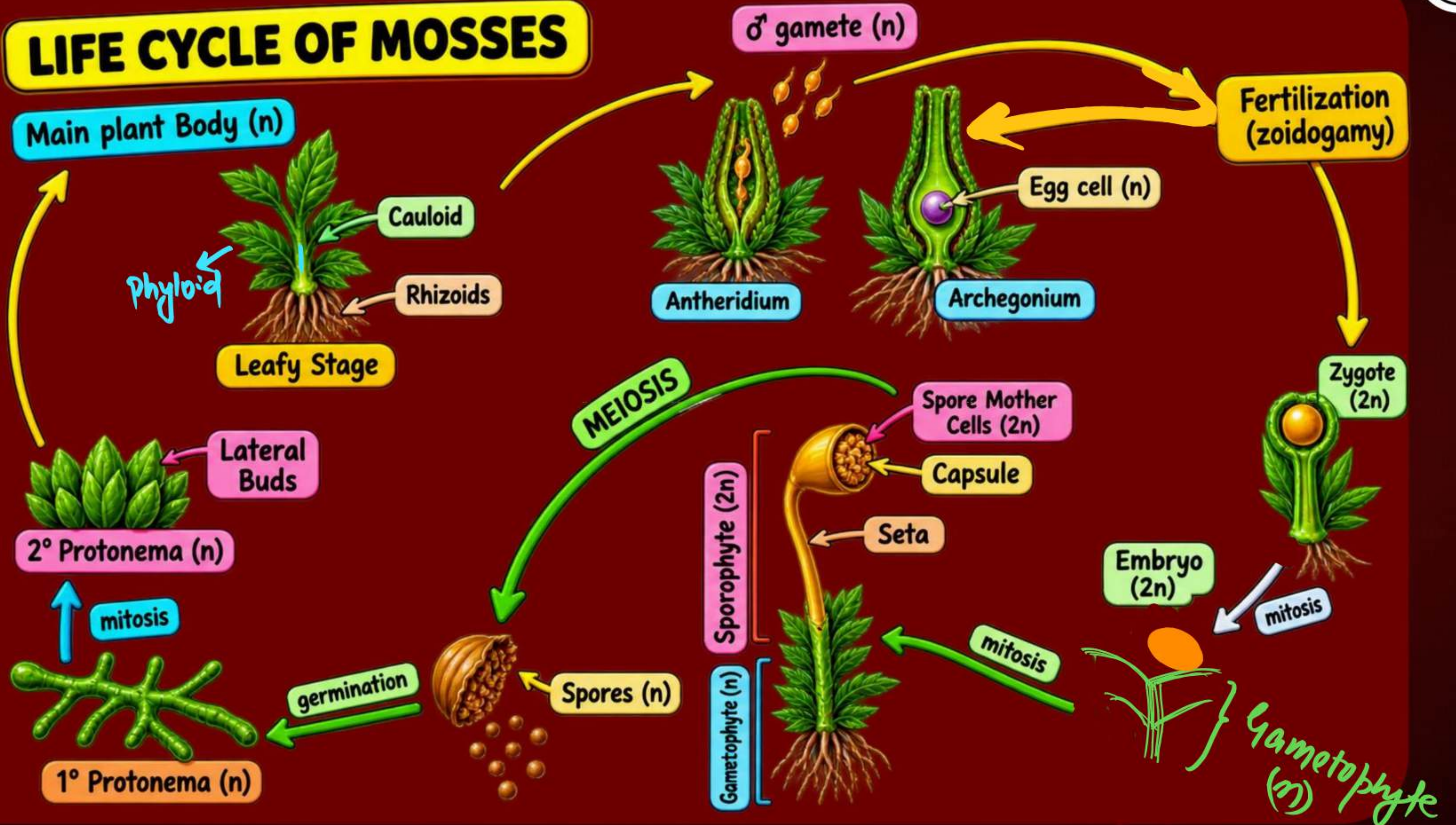
Life cycle — two stages:

a) Protonema — Juvenile Stage. ✓
develops from germination of spore.



b) Leafy Stage — Possess spirally coiled leaf like appendages on Erect Stem like Str.
develops from 2° Protonema as a lateral bud.

LIFE CYCLE OF MOSSES



General Characters of Pteridophytes

Habitat: Cool, damp, shady places and some flourish in sandy-soil.

Pteridophytes — include Ferns and horse tails (Equisetum)

Main plant Body: Sporophyte ($2n$) — produces Spores (n) (By meiosis).

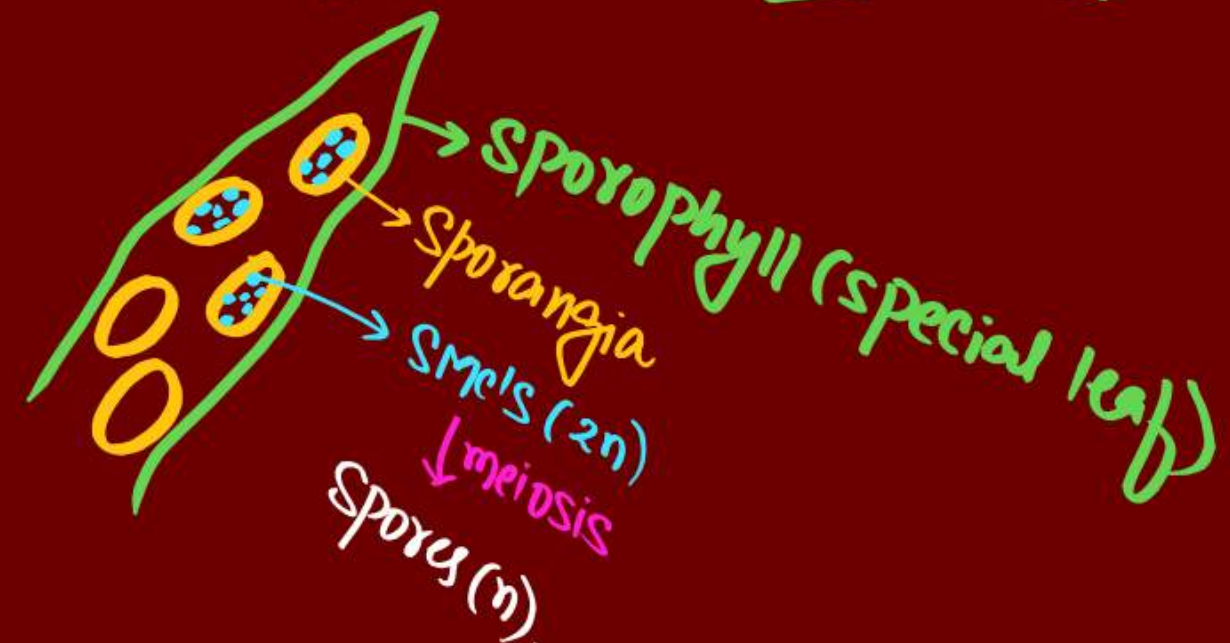
- Vascular system +nt.
- Possess True root, True Stem, True leaves.

Pteridiophytes — first terrestrial vascular plants.

Spores are produced inside sporangia (Borne on special leaf – sporophylls)

Pteridophyta — two types

- ✓ a) Homosporous Pteridophytes.
- ✓ b) Heterosporous Pteridophytes.





FERNS

— Large leaves

Sporophyll

Sporangium

Spore mother cells (2n)

Meiosis

Spore



Homosporous Pteridophytes

- Maximum ✓
- Produce same kind of spore.
- Ex. Ferns, Equistem.

Heterosporous Pteridophytes

Minimum ✓

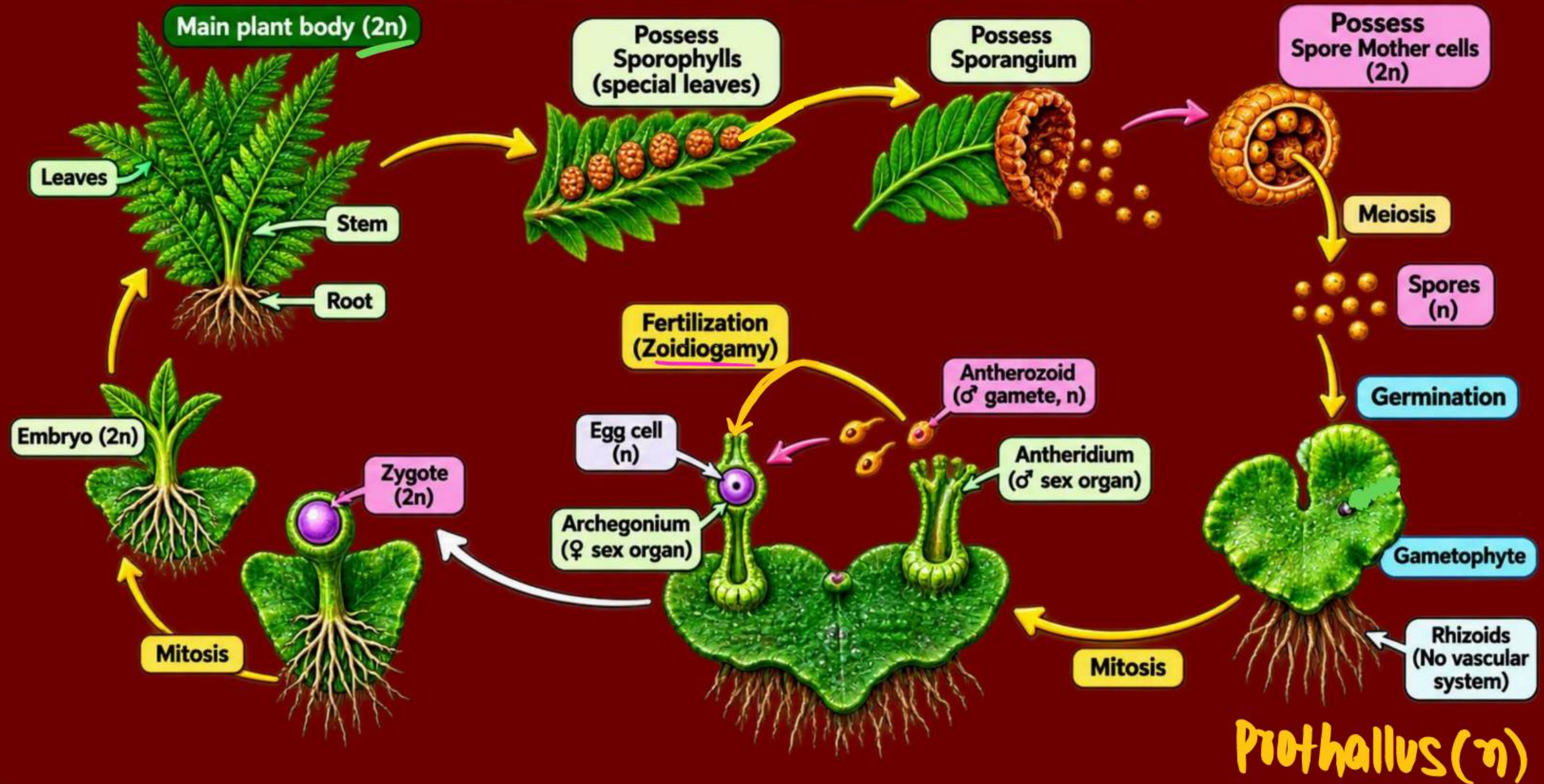
produce two kinds of spores

- ↗ Microspore (small sized)
- ↘ Megaspore (large sized)

Ex. Selaginella.

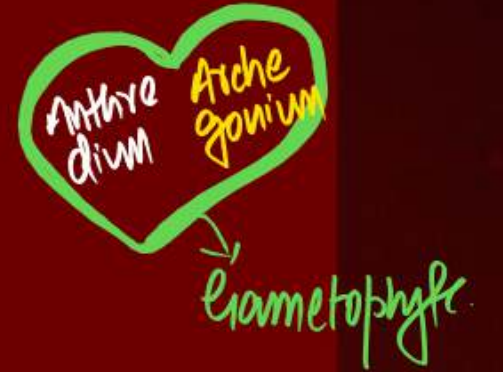
Salvenia.

LIFE CYCLE OF PTERIDOPHYTES



In Homosporous Pteridophyte — Gametophyte is monoecious i.e, both ♂ ♀
Sex organs are ~~+~~nt on same Gametophyte.

In Heterosporous Pteridophytes — Gametophyte is dioecious.



- Microspore
↓
Germinate
↓
Gametophyte (♂ Gametophyte) — bears ♂ sex organ (Antheridium).

Megaspore
↓
Germinate
↓
Female gametophyte (bears ♀ sex organ - Archegonium).
↓
Retained on main plant body

Heterosporous pteridophytes are the precursors of 'seed habit'

Development of seed
presence of seed.

Gymnosperms — Habitat & Main Plant Body

Habitat: Mainly found in cold regions
In India — +nt on Himalayan mountains.

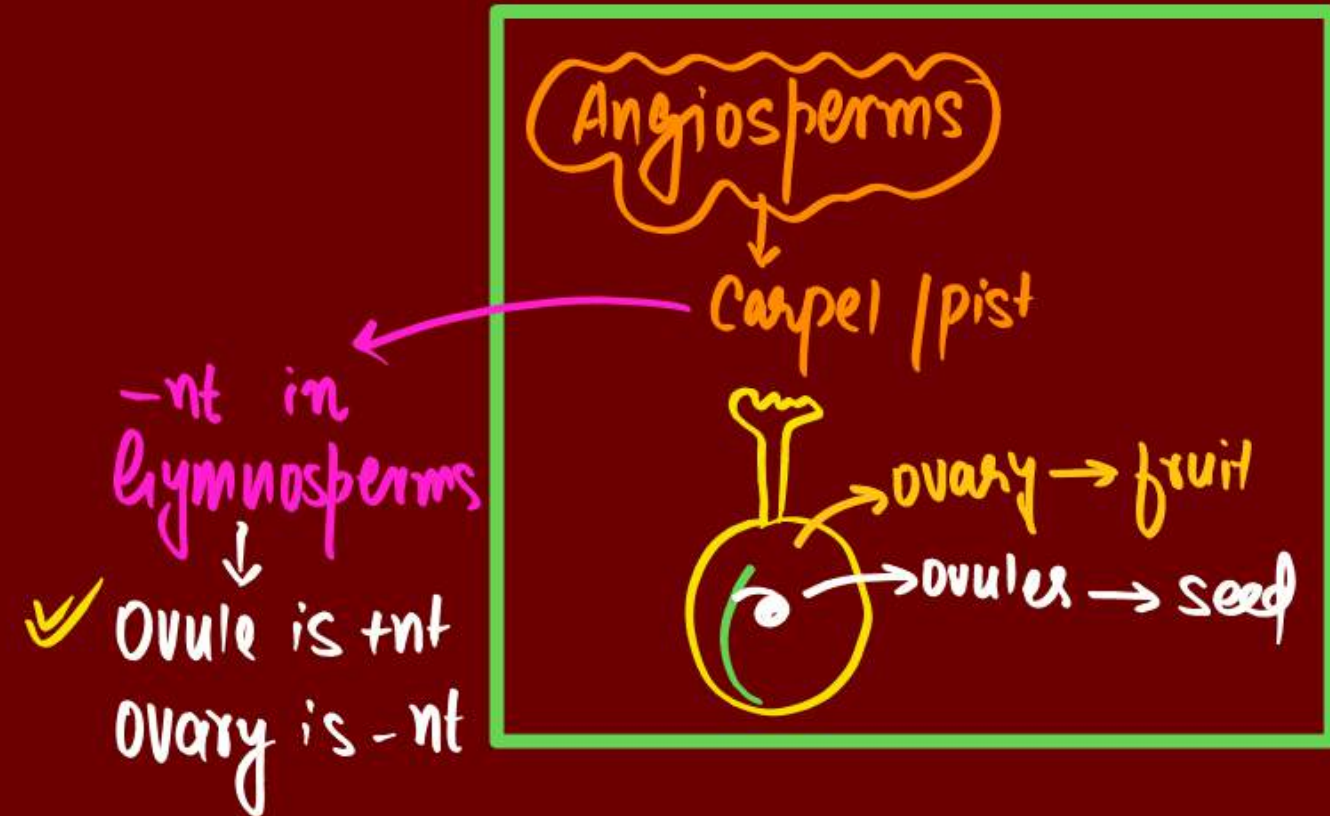
Main Plant Body: Sporophyte (2n)
produces spores by meiosis.

- Vascular System — **PRESENT**.
- Possess —
 - True Root ✓
 - True Stem ✓
 - True leaves ✓

Gymnosperms:

↗ Gymno's — Naked
↘ Sperma — Seed

- Seed is +nt but it remains naked before and after fertilization.
- Ovule is +nt — which forms seed after fertilization.

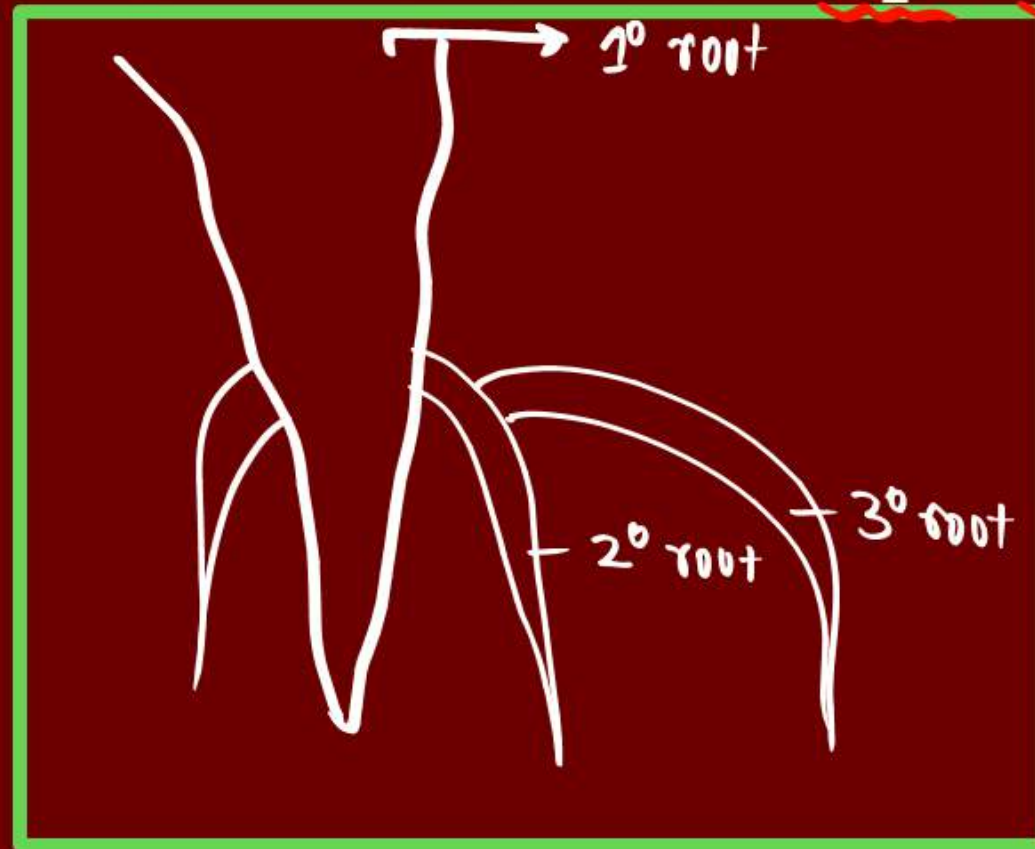


Gymnosperms — Habitat & Main Plant Body

Root — generally tap root.

Pinus — Root in association with fungi (Mycorrhiza)

Cycas — Root in association with N_2 fixing cyanobacteria (coralloid root)



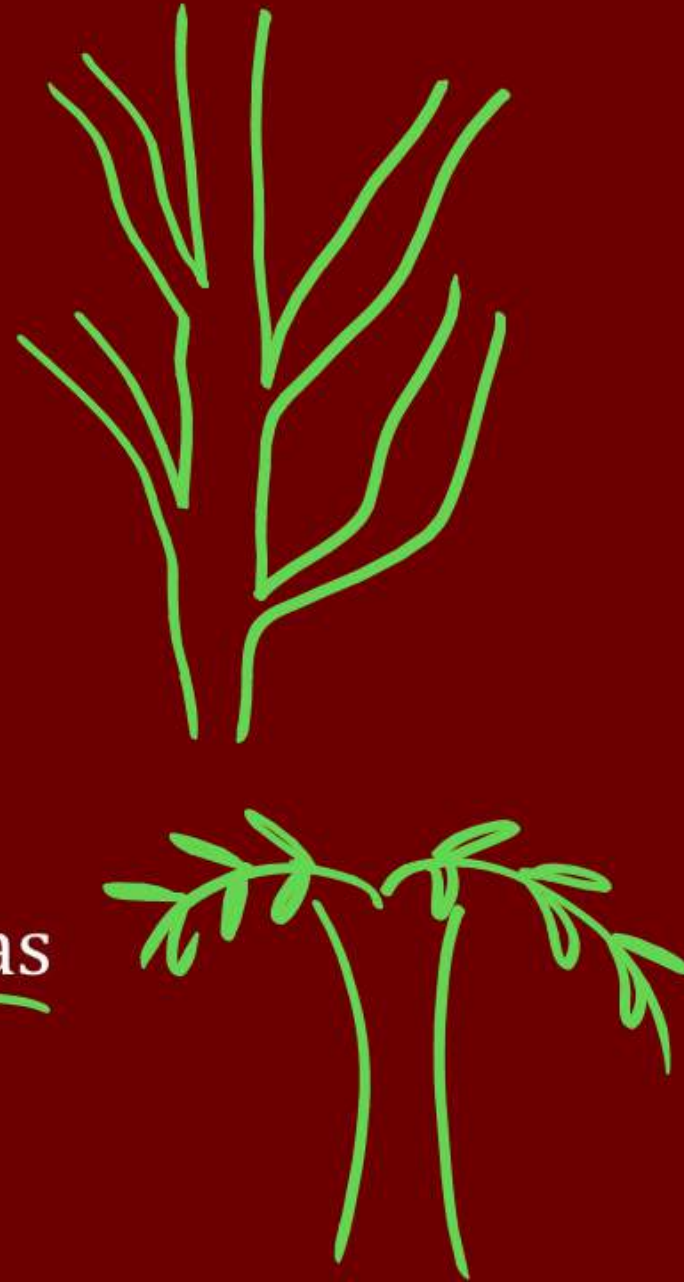
→ 1° root + lateral branches (2°, 3° etc) } Tap root.

Leaves:

- Simple
- Compound
- Possess thick cuticle
- Needle like

Stem:

- Branched — Ex. Pinus
- Unbranched — Ex. Cycas



Gymnosperms — heterosporous ✓

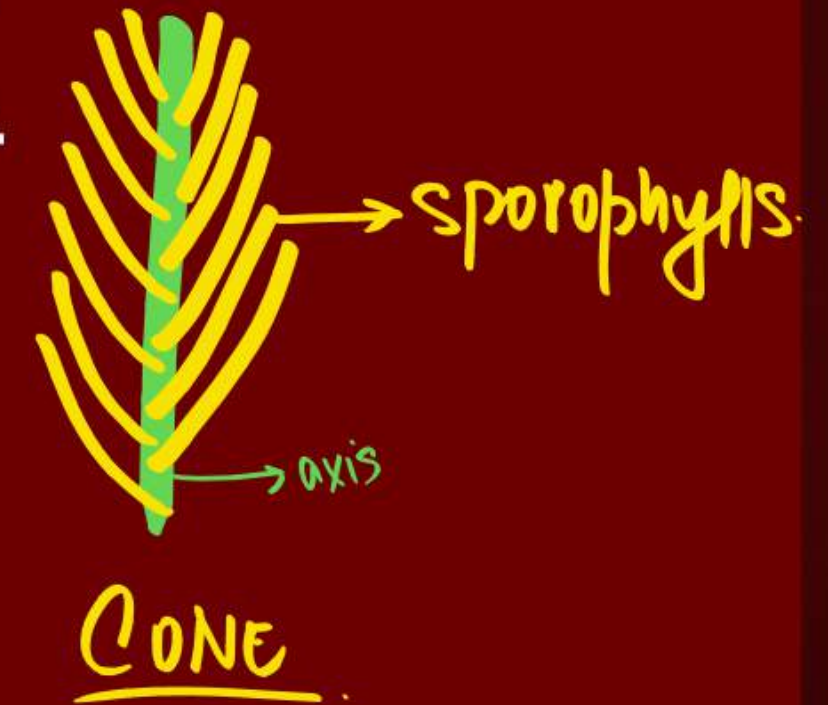
- ↗ microspore (small sized spore)
- ↘ megaspore (large sized spore)

Gymnosperms — possess special leaves called Sporophylls

- Arranged loosely or compactly on axis to form Cone.

CONE:

- ✓ • Male cone
- ✓ • Female cone



Male Cone: (pollen cone) or

Microsporangiate cone.



possess microsporophylls



bears microsporangium



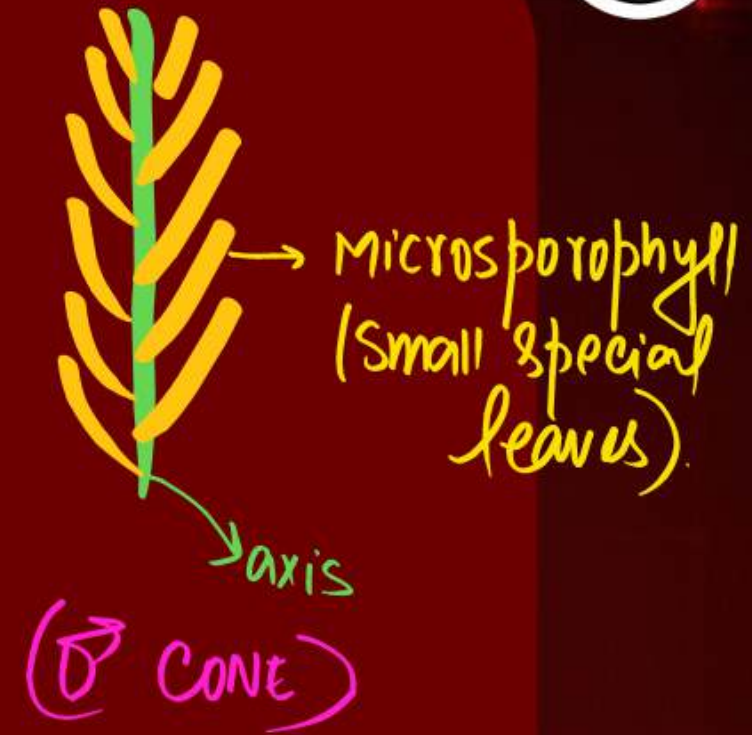
inside it are that microspore mother cells ($2n$)

↓ meiosis

microspores (n)

↓ Development inside microsporangium.

Pollen grain (posses two male gametes)



Female Cone : Seed Cone



FEMALE CONE

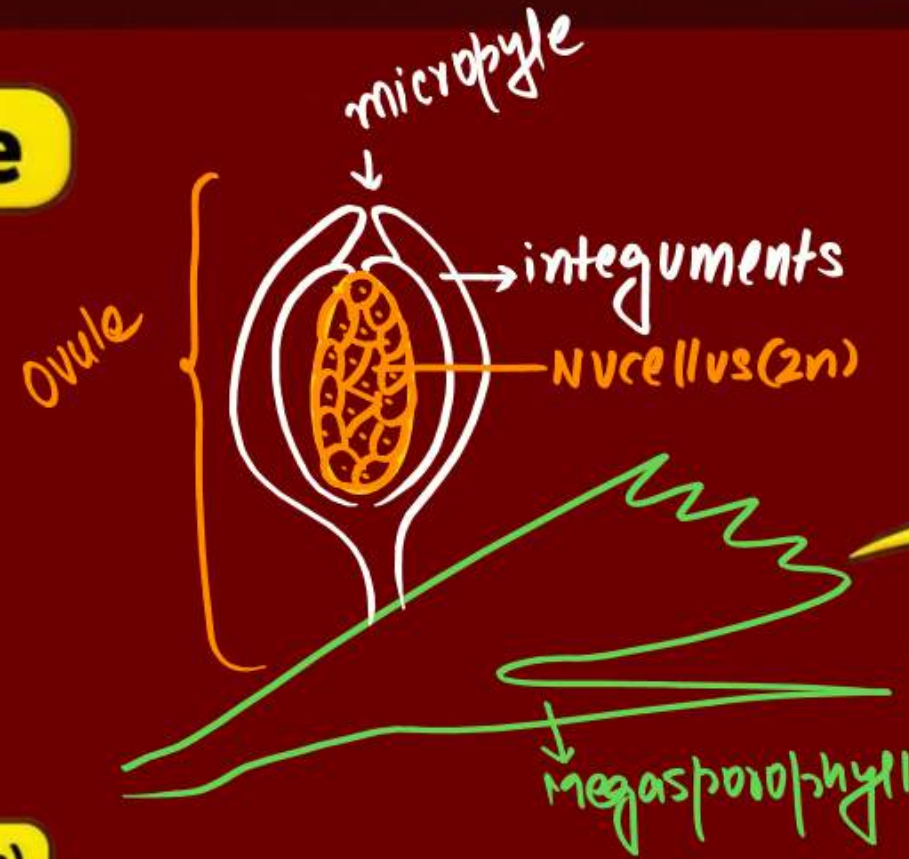
Megasporophylls

large sized sporophylls

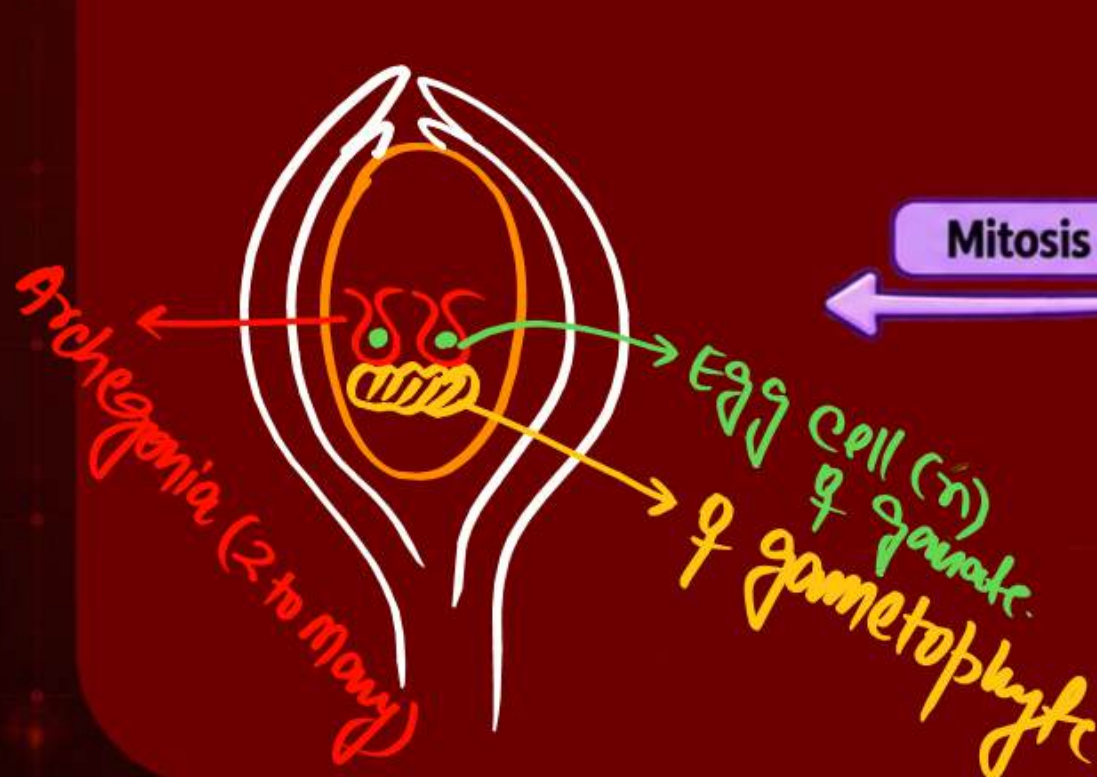


Megasporangiate cone

bears megasporangium (ovule)



One of the cells of nucellus behaves as megaspore mother cell (2n)



Mitosis



Endosperm (n)

OR
♀ gametophyte

3 degenerate

Functional megaspore (n)

4 Haploid megaspores

Develop

Meiosis

MMC (2n)



Pollination: → **By wind (Anemophily)**

Transfer of pollen grains from ♂ cone to micropyle

1 Fertilization:

Pollen grain forms pollen tube

Fertilization by pollen tube –
“Siphonogamy”

Pollen tube → carry ♂ gametes

Integument

Pollen tube (PT)

Integument

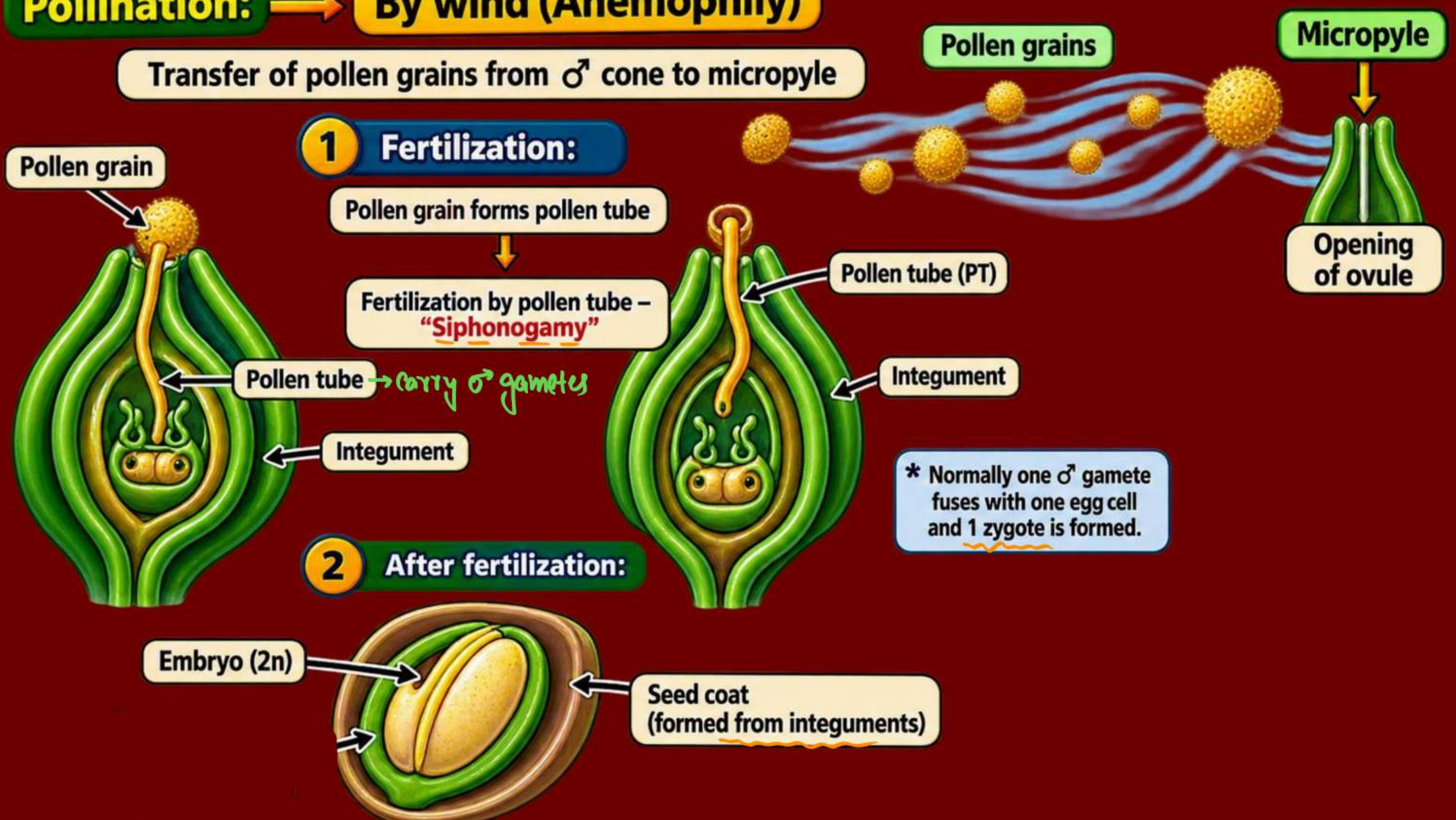
* Normally one ♂ gamete fuses with one egg cell and 1 zygote is formed.

2 After fertilization:

Embryo (2n)

Seed coat
(formed from integuments)

Seed



Polyembryony

- presence of more than 1 embryo in a single seed
- If all the σ gametes fuse with all the Egg cells
- Then many Embryo's are formed in a single seed
- Seen in Gymnosperms

Most IMPORTANT Questions for JKBOSE 2026

Q1. Discuss:

- a) Chlorophyceae ✓
- b) Phaeophyceae ✓
- c) Rhodophyceae ✓

Q2. Important characteristics of:

- a) Bryophyta ✓
- b) Pteridophyta ✓
- c) Gymnosperms ✓

✓ Q3. Why bryophytes are called as amphibians of plant kingdom?

✓ Q4. Why heterosporous pteridophytes are called as precursors of seed habit?



Thank
You